

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Year: 2017

Subject: 341131 Electrical Drawing I

Section: 5-6

Date: 6 October 2017

Time: 13.00-15.00

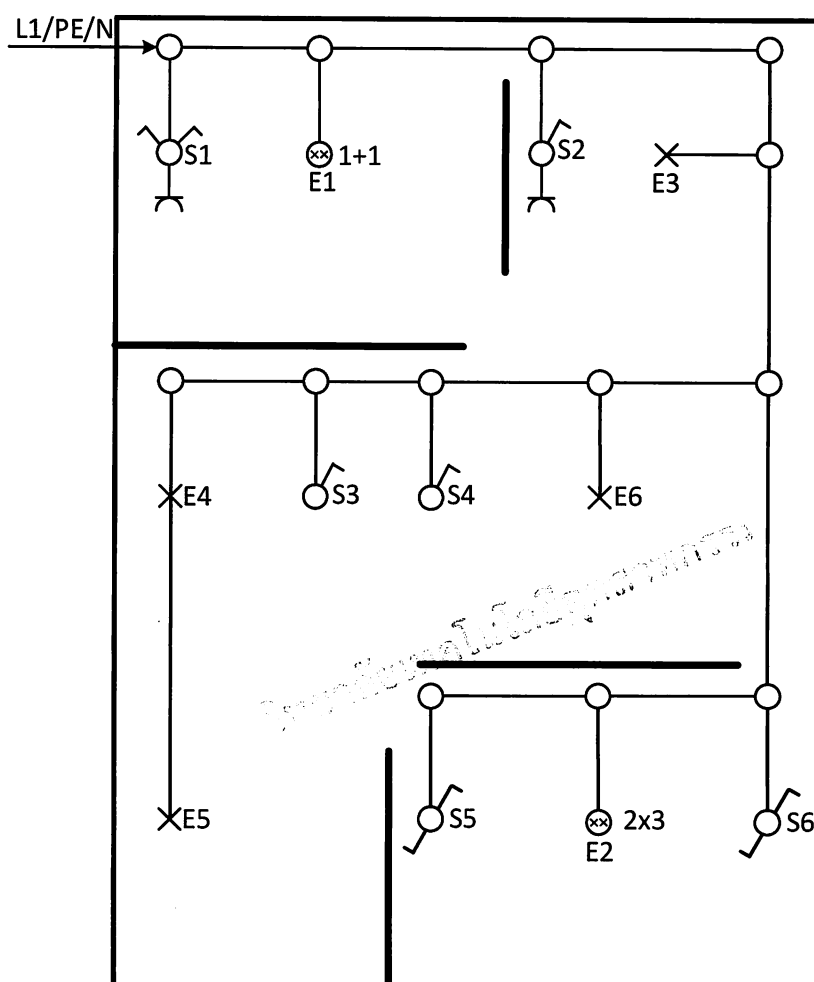
Name..... ID..... EP.....

Instructions:

1. The examination has 5 pages (including this page), 1 question and a total score of 40 points.
 2. Write all your solution and answers on the answer sheet.
 3. This is a closed book examination.
 4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
 5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
 6. You are not allowed to use the restroom during the exam except in case of an emergency.
 7. No documents are allowed to be taken out of the examination room.
 8. Calculator is not allowed in the examination.
 9. Electronic communication devices are NOT allowed in the examination room.
-

**Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.**

1. The floor plan with electrical circuit diagrams is shown below. It include 2 series incandescent lamps (E1 and E2), 4 incandescent lamps (E3, E4, E5 and E6), 1 series switch (S1), 3 switches (S2, S3 and S4), 2 two-way switches (S5 and S6) and 2 plug sockets. Please draw the wiring, one-line and control diagrams. (40 points).



Remarks:

- E1 is controlled by S1.
- E2 is controlled by S5 and S6.
- E3 is controlled by S2.
- E4 and E5 that are connected in parallel are controlled by S3.
- E6 is controlled by S4.
- 2 plug sockets are independent each other.

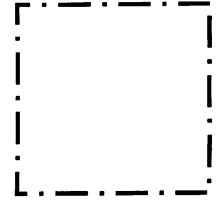
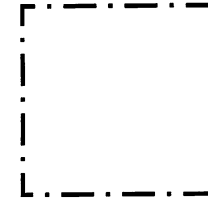
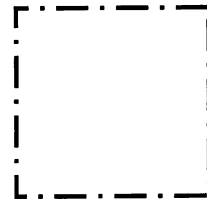
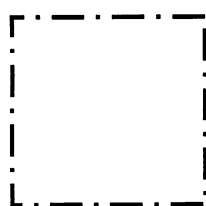
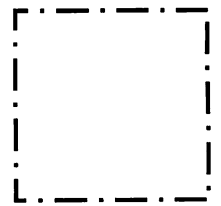
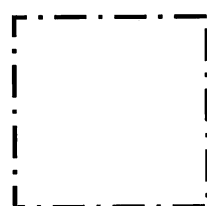
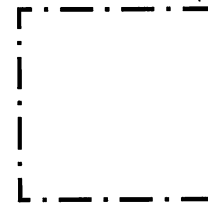
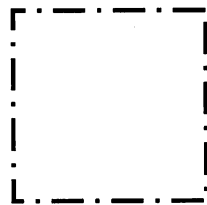
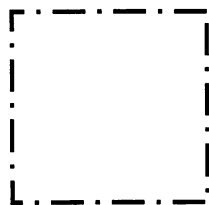
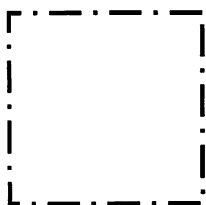
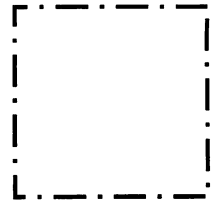
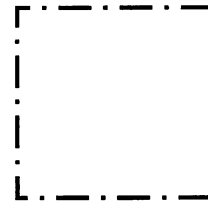
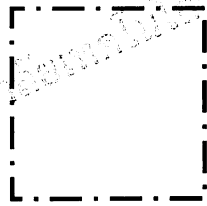
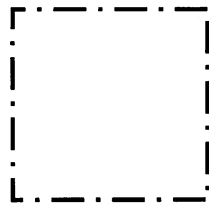
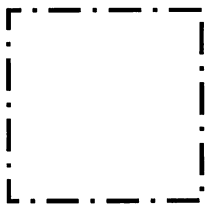
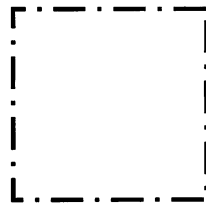
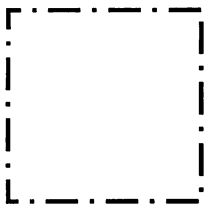
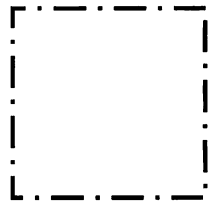
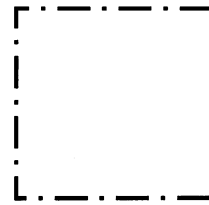
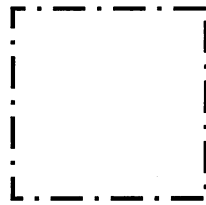
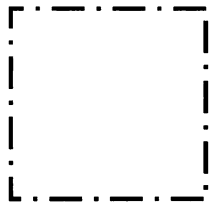
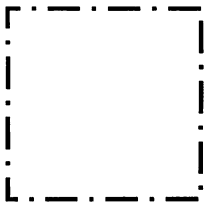
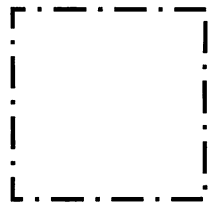
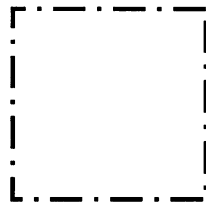
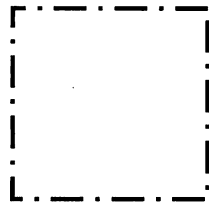
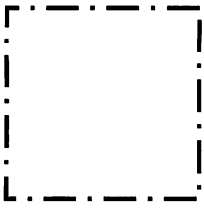
Answer Sheet

One-line diagram (10 points) Name.....ID.....

เลขที่นั่งสอบ

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Wiring diagram (20 points) Name..... ID.....



Control diagram (10 points) Name.....ID.....

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Assoc.Prof.Dr.Rattanakorn Phadungthin

Asst.Prof.Dr.Pinanta Chatwattana